

Renormalized Branch Amplitudes for the Square-Billiard Abel Trace

Route-A structural certificate C162

Abstract

For the Dirichlet half-wave trace W_D of the unit square, we prove that every nonzero source lattice shell admits a canonical full-trace normalization. At $t_N = 2\sqrt{N}$, $\epsilon^{3/2}W_D(\epsilon - it_N)$ tends to $e^{i\pi/4}r_2^{\text{src}}(N)/(8\pi N^{1/4})$; negative time gives its complex conjugate. This advances the earlier branch-location theorem by extracting a finite coefficient from the complete trace, including times where the boundary subtraction has a coincident simple pole. The proof fixes the phase with the principal branch, splits off finitely many shells, and uniformly dominates the infinite remainder by the summable two-dimensional $|m|^{-3}$ tail. Exact shells through $N = 800$ and high-precision convergence rows are regression sentinels, not the proof. The coefficient is aggregate clean-family multiplicity, not an isolated stability amplitude or target divisor law, and no arithmetic Euler or Hilbert–Pólya claim is made.

Keywords: square billiard; Abel half-wave trace; clean family; branch singularity; boundary normalization; Poisson summation.

中文摘要：本文为单位正方形狄利克雷半波迹的每个非零源格点壳层建立规范的全迹边界归一化。在 $t_N = 2\sqrt{N}$ 处， $\epsilon^{3/2}W_D(\epsilon - it_N)$ 的极限为 $e^{i\pi/4}r_2^{\text{src}}(N)/(8\pi N^{1/4})$ ，负时间极限为其复共轭。该结论不仅定位分支点，而且从完整迹中提取有限系数；即便边界扣除项在同一时刻有简单极点，归一化后该极点仍消失。证明以主值分支固定相位，再分离有限壳层，并以二维 $|m|^{-3}$ 可求和尾项统一控制无限余项。直到 $N = 800$ 的精确壳层与高精度数值仅作回归哨兵。所得系数表示光滑轨道族总重数，不是孤立轨道稳定性振幅，也不推出目标除数律、欧拉数据或希尔伯特波利亚算子。

关键词：正方形台球；阿贝尔半波迹；光滑轨道族；分支奇点；边界归一化；泊松求和。

1 Exact source trace

For $\Re s > 0$, the unit-square Dirichlet trace is

$$W_D(s) = \text{Tr } e^{-s\sqrt{\Delta_D}} = \sum_{j,k \geq 1} e^{-\pi s \sqrt{j^2 + k^2}}.$$

C157 established the exact principal-branch Poisson representation

$$W_D(s) = \frac{s}{2\pi} \sum_{m \in \mathbb{Z}^2} (s^2 + 4|m|^2)^{-3/2} - \frac{1}{4} - \frac{1}{e^{\pi s} - 1}. \quad (1)$$

The dual series is locally normally convergent in the right half-plane.

2 Canonical full-trace limit

Let $N \geq 1$, $t_N = 2\sqrt{N}$, and $r_2^{\text{src}}(N) = \#\{m \in \mathbb{Z}^2 : |m|^2 = N\}$. Then

$$\lim_{\epsilon \downarrow 0} \epsilon^{3/2} W_D(\epsilon - it_N) = \frac{e^{i\pi/4} r_2^{\text{src}}(N)}{8\pi N^{1/4}}. \quad (2)$$

At $-t_N$ the limit is the complex conjugate. Indeed the spectral coefficients are real, so $W_D(\bar{s}) = \overline{W_D(s)}$ throughout the right half-plane; the principal dual representation in (1) has the same conjugation symmetry.

For a matching vector, with $s = \epsilon - it_N$, $s^2 + 4N = \epsilon(\epsilon - 2it_N)$. Its normalized contribution tends to

$$\frac{-it_N}{2\pi} (-2it_N)^{-3/2} = \frac{e^{i\pi/4}}{8\pi N^{1/4}}, \quad (3)$$

where the phase is fixed by the principal power. Multiplication by the number of matching vectors gives the right side of (2).

The infinite remainder requires a uniform argument. Fix t_N and $0 < \epsilon \leq 1$. Choose R so that $4|m|^2 \geq 2(t_N^2 + 1)$ for $|m| \geq R$. Then

$$|s^2 + 4|m|^2| \geq 4|m|^2 - |s|^2 \geq 2|m|^2,$$

so the tail is dominated uniformly by a constant times $\sum_{m \neq 0} |m|^{-3} < \infty$. The finitely many nonmatching shells stay bounded. Dominated convergence and $\epsilon^{3/2}$ kill all these terms. If t_N is even, the subtraction term in (1) can

have a simple pole, but its normalized size is only $O(\epsilon^{1/2})$. More precisely, this coincidence means $N = k^2$ and $t_N = 2k$, whence

$$-\frac{1}{e^{\pi(\epsilon - i2k)} - 1} = -\frac{1}{\pi\epsilon} + \frac{1}{2} + O(\epsilon).$$

After multiplication by $\epsilon^{3/2}$ the pole is $-\epsilon^{1/2}/\pi + O(\epsilon^{3/2})$ and vanishes. The Weyl zero-mode is bounded at $t_N \neq 0$, and the constant term is also killed. This proves a limit for the full trace, not just one summand.

3 Certificate and boundary

Exact reconstruction through $N = 800$ finds 270 occupied shells and 2520 nonzero lattice vectors; 28 occupied square norms coincide with simple-pole times. The shell $N = 65$ retains the first four ordered positive primitive directions. Local high-precision rows at $N = 1, 2, 5, 13, 65$ check convergence but are not the proof. For each shell, the exact ledger separately records the four axis vectors when N is a square and all nonaxis sign lifts; gcd reduction of every nonaxis vector retains its primitive direction and repetition. Their total is exactly $r_2^{\text{src}}(N)$ in (2), so no multiplicity is inferred from floating data.

The strict tuple is (A1_WEAK, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION): $\sqrt{\Delta_D}$ is a natural self-adjoint source operator, but the coefficient aggregates positive-dimensional clean families. We claim no isolated stability determinant, target trace/divisor/counting law, arithmetic local/Euler factor, root number, automorphy, Hilbert–Pólya construction, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.

Declarations

Data availability. All claim-bearing data and code are included; no external dataset is used. **Ethics.** No human participants, animals, or sensitive personal data are involved. **Author contributions/CRedit.** Anonymous technical certificate; Conceptualization, Formal analysis, Software, Validation, and Writing are recorded at package level. **Competing interests.** None known. **Funding.** No external funding is reported. **AI-use disclosure.** An AI coding assistant supported derivation planning, drafting, and code review. Packaged independent checks validate quantitative claims; the assistant was not treated as an external peer reviewer.